REVIEW ACTIVITIES: ARMY, MD
NAVY, AS, OS, YD

USER ACTIVITIES:

PART NUMBER	RATED DISTILLATION LITERS (GAL)/HOUR	MAXIMUM OVERALL HEIGHT cm (in.)
MS 36196-1	3.8 (1)	163 (64)
MS 36196-2	11.4 (3)	89 (35)
MS 36196-3	7.6 (2)	173 (68)
MS 36196-4	19.0 (5)	196 (77)

MATERIAL: The apparatus shall be made of materials which meet the design, finish and performance requirements of this standard and the chemical requirements of their respective specifications, when tested as follows:

Dielectric Test. A slowly rising potential, from 0 to 1240 volts, 60 Hz shall be applied between live and dead metal parts. This potential shall be maintained at maximum for one minute. There shall be no evidence of breakdown.

Material Specifications. Copper sheet, hard, cold rolled, QQ-C-576; nickel copper alloy, class A, cold rolled satin finish, QQ-N-281; corrosion resisting steel, class 302 or 304, finish 4, QQ-S-766; brass plate, composition I, condition quarter hard, QQ-B-613; copper tubing, type K, WW-T-799; tin, QQ-T-371; bronze castings, alloy 85, QQ-C-390.

DESIGN: The distilling apparatus shall be of such design as to perform the intended function and shall consist of an evaporator with heating device, condenser, constant level device, deconcentrator, a constant bleeder device, and other equipment necessary for the proper function. The evaporator and condenser shall be connected by means of a vapor pipe. The evaporator shall be heated either by electricity or steam as specified in the invitation for bids.

The electrical heating element shall be the immersion type of 110 or 220 volts, AC or DC.

The steam heated system shall operate between 40 and 60 pounds pressure for satisfactory performance.

FINISH: All metal surface, except stainless steel, which come into contact with vapors or distillate shall have a coating of pure block tin, 0.0001 inch minimum thickness. All other surfaces shall resist corrosion.

PERFORMANCE: The apparatus shall produce distilled water conforming to the requirements of the United States Pharmacopeia when tested as specified in Federal Specification, GG-D-465A, paragraph 4.4.1. Water distillation rate shall be not less than the rate specified in the table.

(A) ENTIRE STANDARD REVISED

APPROVED 23 June 1961 REVISED (A) 7 February 1980

This standard has been approved by the Department of Defense and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

P.A.	EA	TITLE	MILITARY STANDARD
Other Cust	MS 99	DISTILLING APPARATUS, LABORATORY Water, electrically and steam heated	MS 36196
Procurement Specification GG-D-465A		SUPERSEDES:	SHEET 1 OF 2

DD FORM 1 SEP 63 672 (Coordinated) EDITION OF 1 SEP 57 MAY BE USED.

Project Number 6640-0780

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NOTES:
1. Voltage should be specified in the invitation for bids. Automatic cut-off used with 110 volts are not interchangeable with automatic cut-off used with 220 volts.
2. Items that vary in design but which otherwise conform to requirements herein are considered one interchangeable military standard item.
3. Referenced documents shall be of the issue in effect on date of invitations for bids.

REVISED

APPROVED

This standard has been approved by the Department of Defense and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

P.A.	TITLE	MILITARY STANDARD
Other Cust	DISTILLING APPARATUS, LABORATORY Water, electrically and steam heated	MS36196
Procurement Specification	SUPERSEDES:	SHEET 2 OF 2

DD FORM 1 SEP 68 672 (Coordinated) EDITION OF 1 SEP 67 MAY BE USED.

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